

Confid.AI

Software Testing Analysis Report

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This report presents a structured software testing analysis for a selected section of the Confid.AI backend. The section chosen is the **confidence label assignment block** inside the `submit_query()` endpoint (`backend/api/endpoints.py`). This block was selected because it contains clearly defined branches and compound predicates, and it directly controls the confidence label displayed to every user for each AI-generated answer.

1. Code Section Under Test

File: `backend/api/endpoints.py` — Function: `submit_query()`

```
# Thresholds (config.py):
#   HIGH_CONFIDENCE_THRESHOLD   = 0.8
#   MEDIUM_CONFIDENCE_THRESHOLD = 0.5

# Condition A
if not retrieved_passages:
    confidence_score = 0.0
    confidence_label = "Unverified - No Data"
    explanation      = "No documents found."
    citations         = []
    score_breakdown  = { all zeros }
else:
    confidence_score, explanation,
    citations, score_breakdown =
        scoring_service.compute_confidence_score(
            answer, question, retrieved_passages
        )
    # Condition B
    if confidence_score >= 0.8:
        confidence_label = "High - Verified"
    # Condition C
    elif confidence_score >= 0.5:
        confidence_label = "Medium - Verified"
    else:
        confidence_label = "Low - Unverified"
```

2. Code Structure Analysis

2.1 Basic Conditions

ID	Expression	Evaluated When	Possible Values
A	<code>not retrieved_passages</code>	Always on entry	True / False
B	<code>confidence_score >= 0.8</code>	Only when A = False	True / False
C	<code>confidence_score >= 0.5</code>	Only when A = False, B = False	True / False

2.2 Statements

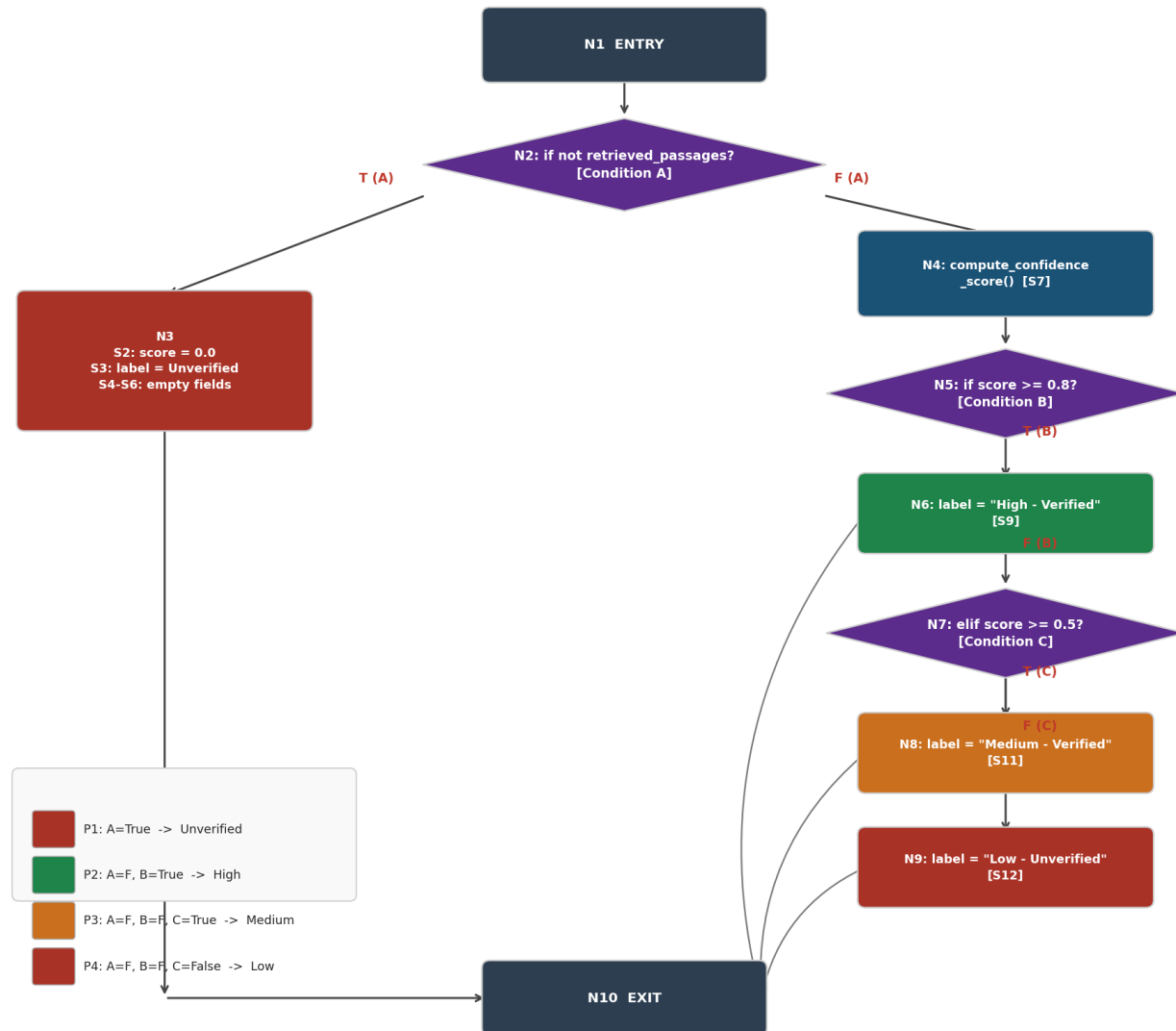
ID	Statement	Paths
S1	if not retrieved_passages (branch check)	All
S2	confidence_score = 0.0	P1
S3	confidence_label = 'Unverified - No Data'	P1
S4	explanation = 'No documents found.'	P1
S5	citations = []	P1
S6	score_breakdown = { all zeros }	P1
S7	scoring_service.compute_confidence_score(...)	P2, P3, P4
S8	if confidence_score >= 0.8	P2, P3, P4
S9	confidence_label = 'High - Verified'	P2
S10	elif confidence_score >= 0.5	P3, P4
S11	confidence_label = 'Medium - Verified'	P3
S12	confidence_label = 'Low - Unverified'	P4

Total: 12 statements · 3 basic conditions · 6 branches · 4 executable paths

3. Control Flow Node Diagram

Rectangles represent process or terminal nodes. Diamonds represent decision nodes. Arrows are labelled T (True) or F (False). The four coloured nodes correspond to the four distinct execution paths.

Control Flow Node Diagram — Confidence Label Assignment Block



3.1 Path Descriptions

Path	Node Sequence	Condition Values	Label Outcome
P1	N1 -> N2 -> N3 -> N10	A = True	Unverified - No Data
P2	N1 -> N2 -> N4 -> N5 -> N6 -> N10	A = False, B = True	High - Verified
P3	N1 -> N2 -> N4 -> N5 -> N7 -> N8 -> N10	A = False, B = False, C = True	Medium - Verified
P4	N1 -> N2 -> N4 -> N5 -> N7 -> N9 -> N10	A = False, B = False, C = False	Low - Unverified

All 4 paths are feasible. There is no dead code or unreachable branch in this section.

4. Test Suite

Four test cases are defined — one per execution path. Together they satisfy all seven coverage criteria. No test shares preconditions with another.

TC-1: No Documents in Knowledge Base (Path P1)

Preconditions	ChromaDB collection is empty. No PDFs have been uploaded to the system.
Inputs	question = 'What is recursion?' retrieved_passages = [] (empty list)
Condition Values	A = True (not retrieved_passages evaluates to True)
Expected Output	confidence_score = 0.0 confidence_label = 'Unverified - No Data' citations = [] score_breakdown = {consistency:0, semantic:0, completeness:0, precision:0}
Statements Hit	S1, S2, S3, S4, S5, S6

TC-2: Strong Document Match — High Confidence (Path P2)

Preconditions	ChromaDB contains a directly relevant PDF. scoring_service returns 0.87.
Inputs	question = 'Define machine learning' retrieved_passages = [doc1, doc2, doc3]
Condition Values	A = False B = True (0.87 >= 0.8)
Expected Output	confidence_score = 0.87 confidence_label = 'High - Verified' citations = [citation list] explanation = set by scoring_service
Statements Hit	S1, S7, S8, S9

TC-3: Partial Document Match — Medium Confidence (Path P3)

Preconditions	ChromaDB contains a partially relevant PDF. scoring_service returns 0.65.
Inputs	question = 'Explain neural networks' retrieved_passages = [doc1]
Condition Values	A = False B = False (0.65 < 0.8) C = True (0.65 >= 0.5)
Expected Output	confidence_score = 0.65 confidence_label = 'Medium - Verified' citations = [citation list] explanation = set by scoring_service
Statements Hit	S1, S7, S8, S10, S11

TC-4: Off-Topic Query — Low Confidence (Path P4)

Preconditions	ChromaDB contains an unrelated PDF. scoring_service returns 0.30.
Inputs	question = 'What is the capital of France?' retrieved_passages = [doc1]
Condition Values	A = False B = False (0.30 < 0.8) C = False (0.30 < 0.5)
Expected Output	confidence_score = 0.30 confidence_label = 'Low - Unverified' citations = [citation list] explanation = set by scoring_service
Statements Hit	S1, S7, S8, S10, S12

5. Truth Tables

5.1 Basic Conditions Truth Table

Test	A: not passages	B: score >= 0.8	C: score >= 0.5	Label Outcome	Path
TC-1	T	--	--	Unverified - No Data	P1
TC-2	F	T	--	High - Verified	P2
TC-3	F	F	T	Medium - Verified	P3
TC-4	F	F	F	Low - Unverified	P4

'--' means the condition is not evaluated (short-circuit logic).

5.2 Compound Conditions Truth Table

Each predicate in this block is atomic. Compound expressions arise from the if/elif/else chain:

Compound Expression	TC-1	TC-2	TC-3	TC-4	T & F Covered?
A (not retrieved_passages)	T	F	F	F	Yes
A = F AND B (score >= 0.8)	--	T	F	F	Yes
A = F AND NOT B AND C (score >= 0.5)	--	--	T	F	Yes

5.3 MC/DC Independence Pairs

Cond.	Test Pair	What Changes	Other Conditions Fixed	Outcomes Differ?
A	TC-1 vs TC-2	A: T -> F	B = T, C = T (implied by score = 0.87)	Yes: Unverified -> High
B	TC-2 vs TC-3	B: T -> F	A = False, C = True	Yes: High -> Medium
C	TC-3 vs TC-4	C: T -> F	A = False, B = False	Yes: Medium -> Low

6. Coverage Metrics

#	Coverage Type	Formula	Calculation	Score
1	All Path Testing	# paths covered / # total paths	4 covered / 4 total	4/4 = 100%
2	Statement Coverage	# statements covered / # total statements	12 covered / 12 total	12/12 = 100%
3	Segment Coverage	# segments covered / # total segments	12 covered / 12 total	12/12 = 100%
4	Branch Coverage	# executed branches / # total branches	6 taken / 6 total	6/6 = 100%
5	Basic Condition Coverage	# truth values / (2 x # basic conditions)	6 values / (2 x 3 = 6)	6/6 = 100%
6	Compound Condition Coverage	# truth values / (2 x # compound conditions)	6 values / (2 x 3 = 6)	6/6 = 100%
7	MC/DC Coverage	# non-redundant combos / (N+1) required	4 tests / (3+1 = 4) required	4/4 = 100%

Note — Compound Condition Coverage: Each predicate is atomic. Compound expressions are formed by the implicit AND chains in the if/elif/else structure (see Section 5.2). All 3 compound expressions take both True and False values across the 4 test cases, satisfying all 6 required truth values.

Note — MC/DC: With 3 independent conditions (A, B, C), the minimum required test set size is $N+1 = 4$. The 4 test cases each form a unique independence pair (one per condition), confirming each condition independently affects the outcome.

7. Justification for Test Suite

Why this code section?

The confidence label assignment block is the most user-facing decision logic in Confid.AI. It determines what label (High, Medium, Low, or Unverified) is shown for every AI-generated answer, drives the admin flagging system, and triggers the red warning banner. Incorrect labelling would directly mislead users, making this the highest-priority section to test.

Why exactly 4 tests?

The section has exactly 4 feasible execution paths, 3 independent conditions, no loops, and no dead code. One test per path is both the minimum and the maximum needed: it achieves 100% coverage on all seven metrics simultaneously, with no redundant test.

Why MC/DC?

MC/DC confirms that each individual condition independently changes the outcome, ensuring no logic is redundant. It is the gold standard for decision-intensive code and is mandated in safety-critical standards (DO-178C, IEC 61508).

Suggested boundary tests for production use

TC-5: confidence_score = 0.8 exactly (boundary for Condition B). TC-6: confidence_score = 0.5 exactly (boundary for Condition C). TC-7: retrieved_passages = None instead of [] (null-safety check).